BRIGHTLIGHT - DATA ANALYTICS

SQL EXERCISE 2 - Aggregate Functions & Grouping

employees_db

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angele
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisc
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisc

Question 1

1. SELECT COUNT(*) AS total_employees
   FROM employees;

column

total_employees
10

2. SQL - total salary paid to IT Department employees

SELECT SUM(salary) AS total_it_salary
FROM employees
WHERE department = 'IT';

total_it_salary
220000

3.

SQL query - avarage salary for HR employees' department

```
SELECT AVG(salary) AS avr_hr_salary
FROM employees
WHERE department = 'HR';
```

avg_hr_salary
49500

4.

SQL query - find highest and lowest salary in the company

```
SELECT MIN(salary) AS lowest_salary,
       MAX(salary) AS highest_salary
FROM employees;
```

lowest_salary	highest salary
48 000	62 000

5.

SQL - Group employees by Department; display total salary paid in each department.

```
SELECT department, sum(salary) AS total
FROM employees
GROUP BY department;
```

department	total-salary
IT	220000
HR	99000
FINANCE	119000
MARKETING	105000

6.
SQL query count how many employees work in each city

```
SELECT city, COUNT(*) AS employee_count
FROM employee
GROUP BY city;
```

city	employee_count
New York	2
Chicago	3
Los Angeles	2
San Francisco	2
Houston	1

SELECT de

7.
SQL query - group employees by department, calculate the average salary in each department, order the result in descending order of average salary

```
SELECT department, AVG(salary) As Avg
FROM employees
GROUP BY department
ORDER BY avg_salary DESC;
```

department	avg_salary
FINANCE	59500
IT	55000
MARKETING	52500
HR	49500

8.

SQL query - find departments where the total salary paid exceeds 100 000

```
SELECT department, SUM(salary) AS total_salary
FROM employees
GROUP BY department
HAVING SUM(salary) > 100 000;
```

department	total salary
IT	220000
Finance	119 000
Marketing	105 000

9.

SQL query - List cities where more than one employee works, ordered by the number of employees in descending order

```
SELECT city, COUNT(*) AS employee_count
FROM employees
GROUP BY city
HAVING BY COUNT(*) >1
ORDER BY employee_count DESC;
```

City	employee_count
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

10:
SQL query - find the department with the highest average salary

```
SELECT department, AVG(salary) AS AVG
FROM employees
GROUP BY department
ORDER BY average_salary DESC
LIMIT 1;
```

department	avg_salary
Finance	59500